

Khyathi B

☎ 509-348-2972

Data Engineer | Washington, USA

✉ khyathib029@gmail.com

SUMMARY

Proactive Data Engineering with 5+ years of experience spearheading the design, development and optimization of end-to-end scalable data solutions within AWS, Azure, and GCP cloud environments. I bring strong expertise in Big Data processing, real-time data ingestion and warehousing and large-scale data migrations (Teradata to Snowflake). Proven ability in architecting cloud-based data warehousing solutions that improved data retrieval efficiency by 30% and driving operational excellence through CI/CD pipeline implementation, containerization (Docker, Kubernetes), and infrastructure as code (Terraform, CloudFormation), consistently delivering measurable improvements in data processing and analytics capabilities.

WORK EXPERIENCE

AWS Data Engineer | Amazon | Seattle, WA

August 2023 – Present

- Provisioned high-availability **AWS EC2** environments, migrated legacy systems to **AWS**, and automated infrastructure with Terraform templates and modules. Improved deployment speed by 30% through reusable **IaC** patterns.
- Designed, developed, and implemented performant **ETL pipelines** with **PySpark** on **AWS EMR**, orchestrated via **Glue** and **Airflow**, to process **500K+ seller records daily** from Seller Central.
- Authored **AWS Lambda functions (Python)** to orchestrate EMR/Glue Spark jobs, handle S3 event pre/post-processing, and package shared libraries; implemented Hive partitioning, bucketing, joins, and SerDes (REGEX, JSON, Avro).
- Ingested real-time weblogs using **Kinesis** and processed them with **Spark Streaming**. Applied data quality checks to flag pass/bad records, improving onboarding dataset accuracy, and designed real-time **data lineage** tracking with **AWS Glue Crawlers** and **metadata tagging** to strengthen governance and auditability.
- Implemented **automated migration pipelines** to AWS and **Snowflake**, ensuring smooth and reliable cutovers with minimal downtime.
- Partnered with product and analytics teams to publish curated datasets and **develop KPIs/dashboards in QuickSight**, reducing query execution time by **30%** for business-critical reports.
- Integrated **Kubernetes (EKS)** with AWS services to improve scalability and manage containerized data services efficiently.
- Applied **Pandas** for advanced data cleaning and transformation of raw onboarding datasets from CSV, Excel and JSON.
- Managed **IAM** roles and data access controls, ensuring compliance with internal security and governance standards.
- Led a cost optimization initiative by **restructuring** glue jobs and reducing redundant data scans in S3, saving **20%** in monthly AWS costs.
- Conducted **incident response** and root cause analysis for critical pipeline failures, implementing proactive monitoring with **CloudWatch** and **SNS alerts**.
- Orchestrated **end-to-end workflows** by integrating **Control-M with AWS services**. This ensured SLA compliance across batch and streaming pipelines

Azure Data Engineer | Starbucks | Seattle, WA

August 2022 – July 2023

- Engineered and operationalized **ETL/ELT pipelines for multi-terabyte** datasets from ERP, IoT, and third-party APIs into a Lakehouse architecture (ADLS Gen2 and Synapse), consistently maintaining a pipeline SLA of over **99%**.
- Developed optimized **Delta Lake tables** within the **Databricks Metastore**, leveraging **partitioning** and **bucketing** to significantly accelerate analyst query performance and improve data accessibility.
- Implemented real-time streaming with **Kafka** and **Spark Streaming** on Databricks, achieving end-to-end latency under 5 minutes.
- Optimized **Spark jobs in Databricks** through joins, caching, and file-size tuning, reducing runtime by 40% and corresponding decrease in compute costs.
- Integrated **Azure Data Factory** with **Blob Storage** and **Databricks**, landing curated datasets in **ADLS Gen2** and publishing to **Synapse**.
- Improved refresh cadence from T+1 to **hourly updates**, automating **90% of batch pipelines** with parameterized ADF workflows.
- Developed **T-SQL stored procedures, functions and triggers** in Synapse and SQL Server to remove manual reconciliation, cutting query execution time by 30%.
- Built and maintained Docker containers on **Azure Kubernetes Service (AKS)** for supporting services. Established CI/CD pipelines using **Bitbucket** and **Bamboo**, improving deployment speed by 40% and reducing manual intervention by 70%.
- Governed data reliability and quality by implementing checks for row counts, schema drift, and null values within **ADF** and **Synapse** pipelines.
- Enhanced pipeline performance and reporting by leveraging partitioning, PolyBase, CTAS, and materialized views.
- Developed reusable PySpark libraries to standardise data quality checks across multiple ingestion pipelines.
- Documented and standardized **Elasticsearch** usage patterns, reducing query-related incidents by 40% and optimizing monitoring for pipeline health.
- Developed **Power BI** dashboards on Synapse datasets, delivering actionable, store-level insights that empowered business stakeholders to drive operational decision-making.
- Built internal tools including **REST APIs in Python** to expose curated datasets and a **PyQt UI** to manage customer and policy records.

- Developed a scalable data platform by migrating legacy **PySpark RDD** workloads to modern **DataFrames** and **Spark SQL**, reducing batch runtimes by ~60%.
- Built ELT pipelines to deliver modeled datasets in **BigQuery** (staged in Google Cloud Storage) and published to **Google Data Studio dashboards**.
- Designed and implemented real-time streaming processes using **Pub/Sub** and **Dataflow** (Apache Beam), integrating with **Dataproc/HBase** for low-latency persistence and landing historical data in **GCS**.
- Orchestrated complex **batch** and **streaming** ETL workflows using **Apache Airflow** on GCP, incorporating robust scheduling, retries, and SLA monitoring.
- Designed and optimized a **Cassandra schema** for high-throughput, time-series IoT data, supporting **~500K writes/sec** with strategic partition key and **Time-to-Live** implementations.
- Managed **containerized** data services and internal APIs using **Google Kubernetes Engine**, ensuring high availability and scalability.
- Configured and secured GCP **networking** (VPC, IAM, load balancing), leveraging **Cloud Shell** for rapid operational tasks and troubleshooting.
- Integrated hybrid cloud data ecosystems by automating AWS **EMR** cluster spin-ups (via **Boto3**), achieving a ~70% faster launch time and enabling cross-cloud data flows.
- Enhanced data quality and reporting accuracy by integrating outputs with downstream SQL Server databases via **JDBC** for post-processing and reporting pipelines. Improved pipeline reliability by implementing robust monitoring, alerting, and error-handling.

- Implemented a modern data pipeline by migrating complex ETL logic from an **Oracle** data warehouse to PySpark, reducing nightly batch processing time by approximately **30%**.
- Built robust batch and real-time data pipelines using Spark as the core engine, orchestrated via Azure Data Factory with advanced features like **triggers**, **Integration Runtimes**, and **Key Vault integration**.
- Modernized data ingestion by migrating processing from **HDFS** and **HBase** into Spark **DataFrames** and Spark **SQL**, significantly improving code maintainability and execution performance.
- Developed a real-time streaming platform using Kafka and Spark Streaming, with robust **checkpointing** and **idempotent** writes to achieve sub-5-minute end-to-end latency.
- Optimized and maintained legacy **Hive** and **Pig** scripts by implementing partitioning and query tuning, resulting in a ~25% improvement in analyst query times.
- Enhanced data quality and downstream reporting by writing advanced SQL (**T-SQL**, **ANSI SQL**) for validation and reconciliation, which decreased recurring report runtime by **~20%**.
- Promoted ADF pipelines across environments using **ARM templates** to standardize config and lowered deployment errors. Collaborated across **SDLC** phases to deliver releases on schedule and within data SLAs.

TECHNICAL SKILLS

Big Data:	Apache Spark, Kafka, NiFi, Apache Airflow, Flume, Delta Lake, Snowflake, Hadoop, HDFS, Yarn, Hive, HBase, MapReduce, PIG, Sqoop, Oozie, Mahout, Sparklib
Databases:	PostgreSQL, MongoDB , MySQL, DynamoDB, Cassandra , SQL Server, Oracle
Programming:	Python, PySpark, Java, SQL, Shell script(bash, shell), Scala, C
Cloud Technologies:	AWS(EMR, Glue, Lambda, Kinesis, S3, Athena, Redshift, DynamoDB, CloudWatch), Azure(Databricks, Azure Data Factory, Synapse, ADLS Gen2, Azure SQL), GCP(BigQuery, Dataproc, Dataflow/Beam, GCS, Pub/Sub)
Tools:	PyCharm, Eclipse, Visual Studio Code, Docker, Kubernetes, Postman, Jenkins, Terraform, Kibana, SQL Developer, SQL Server Management Studio, TOAD, SQL*Plus, SQL Navigator, Query Analyzer, SQL Assistance
Versioning:	GIT, GitHub, Bitbucket, SVN
Database Modelling:	Dimension Modeling, ER Modeling, Star Schema Modeling, Snowflake Modeling
Methodologies:	Agile(Scrum), Waterfall, SDLC, UML
ML Techniques:	Linear & Logistic Regression, Classification and Regression Trees, Random Forest, Associative rules, NLP, Clustering

EDUCATION

CERTIFICATIONS

- Microsoft Certified:** Azure Data Fundamentals
- AWS Certified:** Data Engineer Associate
- Google Cloud Certified:** Cloud Engineer Associate